TB-001

Optical Performance Under Electrical Load

Why dynamic optical qualification should be part of every heated transparency approval process.

1. Executive Summary

Electrically heated transparencies are qualified today largely with optical methods developed for passive glass. Transmission, haze, and distortion are typically measured with the transparency de-energized and at room temperature. In service, the same transparency spends its most critical hours energized, thermally loaded, and optically dynamic.

This bulletin argues a single engineering principle: if a transparency operates under electrical load, it should be qualified under electrical load.

Engineering principle — static optical qualification characterizes the article as built. Dynamic optical qualification characterizes the article as operated.

2. What Changes When Power Is Applied

- Localized thermal gradients develop between conductors, altering refractive behavior across the aperture.
- Interlayer softening and stress redistribution can produce transient distortion that recovers on cool-down.
- Conductor visibility, diffraction, and glare signature can change with temperature and viewing geometry.
- Bus bar zones and lead exits introduce steeper gradients than the vision area average.
- Repeated power cycling can reveal drift that a single ambient measurement cannot detect.

3. Why Static Data Under-Describes Operational Behavior

Evaluation condition	What it does and does not capture
Ambient, de-energized	Captures as-built optical quality. Does not capture thermally induced distortion or dynamic uniformity.
Soak only, no power	Captures bulk temperature effects. Does not capture conductor-level gradients.
Energized, steady state	Captures operational distortion and uniformity in the condition the operator actually uses.
Energized, cyclic	Captures repeatability, drift, and recovery after power removal.

4. Recommended Practice

- Define the electrical operating envelope before optical acceptance criteria are written.
- Measure optical performance at ambient, at steady-state power, and after recovery.
- Identify critical viewing zones and evaluate them separately from the full aperture average.
- Record synchronized electrical, thermal, and optical data so results are traceable.
- Repeat across power cycles to establish repeatability rather than a single-point result.

5. Conclusion

Dynamic optical qualification does not replace existing standards. It closes the gap between how a heated transparency is inspected and how it is used. HLG-QS-1724 provides the framework HLG applies to this evaluation.

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